

ROADM and AI: a Perfect Symbiosis for Autonomous Optical Networks

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Abstract *This article examines how AI is reshaping optical transport networks by driving architectural innovations to meet surging data center interconnect demands, while simultaneously enabling intelligent, agentic network automation frameworks that optimize operations, reduce costs, and deliver measurable business outcomes. ©2026 The Author(s)*

Introduction

Over the past five years, the telecommunications industry has undergone a profound transformation, largely driven by the rapid growth of the introduction of Artificial Intelligence (AI)-based models [1]-[3]. One of the most significant impacts of this evolution has been a shift in the primary buyers of WDM metro/core (and subsea) network equipment. For nearly three decades, the main customers were Telecommunication Providers (TPs), or Multi-service operators (MSOs) whose essential objective was to interconnect points of presence across and within countries, linking major cities. In recent years, however, AI and Cloud provider – operators of large-scale cloud computing infrastructures – have become increasingly active in this market and are fundamentally reshaping it. According to [4], by 2026, AI and Cloud providers are expected to account for more than 30% of global optical hardware spending, a figure that rises to approximately 60% in the United States alone. As AI and Cloud providers capital investments continue to grow to keep up with the demand for AI, network architectures are evolving to align as closely as possible with their requirements. The primary objective for AI and Cloud providers is to interconnect massive volumes of data between data centers (DC) located within metropolitan areas as well as across different regions [5].

The massive volume of traffic exchanged between DCs is primarily driven by the training of Large Language Models (LLMs) and other AI models that require tens of thousands of GPUs [6]. These GPUs are increasingly deployed across geographically distributed DCs due to physical space limitations and the growing power consumption constraints of individual facilities [7]. As a result, high-capacity, low-latency interconnection between DCs – both within metropolitan areas and across regions – has become a fundamental requirement. This trend further amplifies the demand for scalable and efficient optical

transport networks capable of supporting unprecedented bandwidth growth [7].

While the first transformation concerns the physical layer of optical networks, a third – equally consequential – shift is reshaping how they are operated [8]. Until recently, AI in optical networking referred to the workloads traversing the fiber; it is now also becoming the technology that runs it. Driven by mature deep-learning models and GPU compute, network operations are moving beyond rule-based automation toward agentic systems: software agents that perceive network state, reason over it, and act within closed-loop control [10], [11]. This promises to compress fault-recovery times, improve spectral efficiency through continuous re-optimization, and lower the OpEx that has historically scaled with network size. In the remainder of this article, we examine both faces of this transition: how AI workloads reshape optical transport node architectures, and how AI techniques enable autonomous network management.

The demand for architectural change

The per-fiber capacity of modern data center interconnect (DCI) DWDM systems operating in the C and L bands can reach up to 76.8 Tb/s based on standard channel spacing (e.g. 64 x 150GHz channels of 1.2Tb/s) or more with tighter channels spacing (e.g. 72 x 132.5GHz channels of 119Gbaud 1.2Tb/s) or with the additional 2THz of Super C and Super L [12]. However, when considering DCI links, the deployment of dozens, even hundreds of parallel fibers – referred to as rails – between them is used to increase the capacity. The need for expanded growth of transport fibers between locations hosting the GPU compute resources depicts bottlenecks based on constraints such as rack space, power availability, and the management of many modules within the nodes at each geographically dispersed location. While this depends on the network operator, the challenge resides with the complexity involved in managing networks

designed for AI scale, performance and SLAs. Simplifying optical multilayer management and performance monitoring across transport segments connectivity DCs becomes vital for maintaining the efficiency of GPU cluster interconnects.

AI and Cloud provider network evolution

When the network infrastructure is owned and managed by AI and Cloud providers, IPoDWDM architectures tend to dominate within both metro and regional areas due to factors such as low-power, smaller density and footprint and the higher performance achieved using current 800G coherent pluggable interfaces that can scale to reach distances required between these data-centers [5]. To support this DCI expansion with an increased number of fiber pairs between them – 16 up to 128+ have been documented at OFC2026 [13], [14], inline amplifiers must also become more compact, while Reconfigurable Optical Add/Drop Multiplexers (ROADM) nodes can be significantly simplified, a huge amount of traffic need to be transported between adjacent IP router [15]. This evolution is the rationale behind the latest adoption towards direct-attach ROADMs, multi-rail amplifiers and full-band transponders [14]. Multi-rail amplifiers can amplify up to four fiber pairs within a 1RU chassis across both C and L bands, leveraging shared common equipment—such as control systems, integrated dynamic gain equalizer, optical channel monitors, and optical time-domain reflectometers—rather than treating each fiber pair as an independent line system requiring dedicated hardware and replicated control infrastructure. Full spectrum transponders (FSTs) provide a second evolution on the road to multi-rail. FSTs provide a single transponder card with multiple coherent engines capable of filling an entire band (e.g., 16 x 300GHz 1.6T) or full fiber (e.g., 32 x 300GHz 1.6T) in a compact (1RU or 2RU) device. These FSTs typically integrate line system functions including mux/demux and amplifiers, removing or at simplifying, separate/external line systems at the terminal sites.

Telecommunication Provider managed optical fiber networks

When the optical network infrastructure is operated by TPs, different capacity demands must be considered. For traditional wholesale customers, capacity demands are typically on the order of a few terabits per second between arbitrary nodes in a mesh network configuration. In this case, networks continue to rely on optical bypass capabilities provided by conventional Wavelength Selective Switching (WSS) ROADM architectures. In other business model scenarios, TP customers may include AI and Cloud providers or smaller cloud providers that, for various reasons

– resources, regulation or operational risk - cannot deploy their own fiber infrastructure and instead lease geographical segments or bandwidth from an existing network operator. This model is commonly referred to as a Managed Optical Fiber Network (MOFN) [16], in which a high-capacity optical fiber infrastructure is designed, built, and dedicated to a single AI and Cloud provider but fully operated, managed and maintained by a third-party network operator. In such MOFN models, services require the interconnection of distant nodes using extremely large spectrum allocations. In such a case, either point-to-point fixed DWDM or ROADMs may be used; however, the use of the traditional WSS-based switching within the ROADMs may become inefficient due to the very high number of ports required to connect a large bundle of rails between endpoint locations when considering full spectral band deployment. Moreover, WSS solutions with such port counts are currently not available at an affordable cost. To address this challenge, network operators may introduce optical fiber switching, which can significantly reduce node operational complexity and maximizes time-to-service when customers request rapid deployment of new full-band capacity on a fiber pair while also considering the ability to retain resiliency within the interconnection. In more complex scenarios, nodes may adopt modular architectures, supporting multiple switching granularities—from whole fibers to spectral bands and down to individual wavelengths.

What AI does today in the optical lifecycle (planning and design)

Artificial intelligence is already influencing the tooling used to plan and operate optical networks, with maturity in the planning and design stages. In contemporary planning workflows, data-driven models can be trained to estimate optical impairments with high fidelity when they are provided with essential engineering inputs such as fiber characteristics, candidate paths, and transmission rates [17], [18]. These models can then be used to validate prospective optical connections, and—where margins are insufficient, recommend tuning actions that improve feasibility and performance. In practice, this shifts planning from purely rules-based feasibility checks to a hybrid approach in which physics-aware constraints are complemented by learned mappings from historical deployments and operational observations. Such AI-assisted planning is especially useful during design optimization, where the objective is not merely to find a feasible configuration but to select coherent transmission profiles that maximize capacity while preserving adequate system margins [19]. Domain-specific assistants such as “Optics GPT” [20] have been proposed as examples of how specialized AI models can support

optical design, validation, and troubleshooting across this lifecycle stage, serving as an interface that helps practitioners translate high-level intent (e.g., capacity targets and constraints) into engineering-consistent recommendations. This lifecycle view is commonly framed as a sequence of phases—planning, commissioning, provisioning, assurance, analysis, and optimization—each comprising a set of tasks that are often time-consuming and error-prone when performed manually. Within the planning phase, representative activities include creating sites, assigning interconnect rates, defining span lengths and fiber types, and executing feasibility analyses. The immediate contribution of AI at this stage is so not limited to “prediction” in the abstract; rather, it is an enabling layer that can reduce procedural friction in building and validating candidate designs and can accelerate iteration by highlighting likely infeasibilities earlier in the process.

Why ROADMs are a strong substrate for agentic operation

ROADMs are particularly well suited to underpinning more autonomous operation because they sit at the intersection of programmability and observability in the optical layer. Modern optical line systems increasingly integrate operational features that reduce manual effort (e.g., simplified installation with automated discovery) while expanding the availability of in-situ sensing and diagnostics [10]. The use of ROADMs within the network inclusive of various sensors embedded within them, such as optical power monitors, dynamic gain equalizers, , power profile estimators (PPE) [21] and optical time domain reflectometers (OTDR) make them good candidates for agentic systems. Notably OTDRs have become a standard capability with use cases that include fiber cuts localization, detection of increased fiber loss, and intrusion. ROADMs provide both actuation points that can reconfigure optical connectivity and embedded sensing and telemetry that can ground decision-making in real network state. As ROADM systems increasingly expose both raw and derived optical measurements (including alarm and performance monitoring), the ROADM node becomes a data-rich substrate on which higher-level reasoning and closed-loop optimization can be built.

What do agentic systems add to network lifecycle management

While AI-assisted planning improves design-time decisions, agentic systems extend AI’s role into the operational domain by adding autonomous sequencing, tool use, and verification. A practical framing is a phased evolution from “read” to “write” authority. In an initial phase, the automation stack observes network telemetry, analyses alarms and performance data,

diagnoses candidate root causes, and recommends actions accompanied by explicit reasoning, leaving final approval to a human operator. Digital twins (DT) are a key enabling component for trustworthy closed-loop control. They provide a simulation and verification environment that mirrors real-time state information and allows “what-if” evaluation of candidate actions before changes are applied to production. By evaluating hypothetical futures, DTs can help identify configurations that mitigate conflicting side effects and support multi-objective decision-making. Within an agentic closed loop, DT validation becomes the “test” stage that complements detect–diagnose–decide–act, supporting safer autonomy and more reliable convergence to policy goals [22], [23].

What makes AI for networks trustworthy

Trust is the gating factor for deploying AI in mission-critical network operations. Network-grade AI requires a different posture from general-purpose automation: operators need deterministic outcomes, explanations for why actions were recommended or executed, and accountability when systems act on the network. Key concerns include accuracy and reliability, explainability gaps, hallucination risk in generative systems, and governance of decision authority. As a result, a trustworthy autonomy strategy typically combines guardrails and governance with rigorous operational observability. In practice, this means that AI systems should not only perform actions but also produce auditable artefacts: what inputs were used, what constraints were applied, what reasoning led to the decision, and what rollback or recovery paths are available if outcomes diverge from expectations. This “trust must be earned” principle motivates architectures that emphasize progressive autonomy, explicit validation, and traceable behavior rather than opaque optimization.

Conclusions

The rapid growth of AI is fundamentally transforming optical transport networks across two key dimensions. First, architecturally, the surge in DCI demands is driving the adoption of innovative solutions—including direct-attach ROADMs, multi-rail amplifiers, and full-band transponders—to deliver scalable, high-capacity, low-latency connectivity. Second, operationally, AI-powered agentic systems are revolutionizing network management through autonomous closed-loop frameworks capable of real-time fault detection, self-healing, and SLA optimization. These shifts position AI not only as a traffic driver, but as an enabler of the next generation of intelligent, efficient, and resilient optical networks.

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